STAT606 Assignment Instructions

Find a data set to perform *binary* classification. Follow the tasks below and prepare a presentation of the results. The presentation should be no more than 10 minutes.

1. Give a brief introduction to the data and problem.
2. Familiarize yourself with the data. Fix any inconsistencies (missing values etc.) that may exist (you can use any software for this). How these were fixed should be *briefly* mentioned and motivated in your presentation.
3. Give some basic characteristics of the sample and target variable as well as state what attributes/features were included.
4. You may use existing attributes to create new ones to be used in the models if you think they would be more suitable/informative.
5. Using a 70:30 train vs test set split on the data, apply ***four*** machine learning techniques to classify the target based on the set of attributes in the data, one of which must be a ***decision tree*** method.
6. Comment on the results in terms of any overfitting, underfitting, high variance etc. that may be present. Determine the best performing model according to a measure of your choice. **Substantiate the measure you use**.
7. Provide the top 5 most important features based on your best performing model.

IMPORTANT NOTE: You can source data from your workplace, Kaggle, UCI Machine Learning Repository, OpenML, Machine learning datasets, etc. The target must be BINARY and there must be a range of attributes (categorical and numerical). Please try find a use-case that is relevant/interesting to you and/or your group member. Do not simply go for the easiest dataset (customer churn or bank loan examples).

SUBMISSION INFO:

- You will have a choice to submit a *pre-recording* of your presentation or *present live* online during the course of the day on 18 or 19 May (you will get the option to select your presentation time on a Google sheet on a first come first serve basis closer to the time). Both group members are expected to contribute to the presentation.
- Your **presentation in PDF**, the **R scripts** containing the code for the models fitted (code from the STAT606 must be used, NO AI generated code will be accepted) and the **pre-recording** (if applicable) must be submitted on Moodle (under the *Assessments* topic) by **17h00 on Tuesday 19 May**.